When Delegation Meets Framing: Experimental Evidence on Mass Preferences over Healthcare Spending

Hector Bahamonde*[1], Mikko Niemelä†[2], and Katri Aaltonen‡[3]

[1]Senior Researcher, INVEST Research Flagship Centre, University of Turku, Finland
[2]Professor of Sociology, INVEST Research Flagship Centre, University of Turku, Finland
[3]Academy Research Fellow, INVEST Research Flagship Centre, University of Turku, Finland

February 23, 2026

Abstract

Governments often insulate authority over costly and controversial healthcare funding decisions from day-to-day electoral politics, relying on technical assessments and objective criteria to justify those decisions. Yet public responses to these decisions remain highly sensitive to how identical choices are presented. This article examines whether institutional insulation constrains framing effects on mass preferences, or whether citizens' evaluations of state funding decisions remain elastic to framing even when authority is formally insulated from electoral politics. Drawing on prospect theory, we argue that preferences over healthcare spending are reference-point dependent, such that citizens respond differently when funding decisions are framed as avoiding salient losses versus preserving existing gains. We test this argument using a population-based survey experiment embedded in the 2021 Finnish Medicines Barometer, in which respondents evaluate an identical clinical vignette involving a novel, high-cost cancer medicine with uncertain benefits and are randomly assigned to alternative frames emphasizing last-resort rescue or stewardship of finite collective resources. Holding clinical evidence, costs, and uncertainty constant, we find substantial and systematic framing effects on mass preferences over public funding. Support for funding increases when decisions are framed as loss avoidance and decreases when framed as gain preservation, despite identical policy content. These findings demonstrate that institutional insulation from electoral politics does not neutralize framing effects on mass preferences. Instead, decisions justified through technical criteria remain politically consequential because public communication and interpretive frames shape how citizens evaluate distributive choices. By integrating research on framing with the political economy of delegation, this article explains why distributive healthcare policies remain politically contested even when citizens are not the formal decision-makers and highlights the behavioral foundations of public responses to state action under conditions of risk, scarcity, and institutional insulation.

Abstract length: 258 words.

PLEASE DO NOT CIRCULATE.

Keywords— framing effects; healthcare rationing; delegation; mass preferences; political economy.

*hector.bahamonde@utu.fi; `https://www.hectorbahamonde.com`
†mikko.niemela@utu.fi; `https://www.utu.fi/fi/ihmiset/mikko-niemela`.
‡katri.m.aaltonen@utu.fi; `https://www.utu.fi/fi/ihmiset/katri-aaltonen`

1 Rationing, delegation, framing and mass preferences in the healthcare context

All health systems must set priorities in order to allocate scarce fiscal, personnel, and medical resources. Rationing occurs when potentially beneficial interventions are withheld from individuals (Scheunemann and White 2011). It is inherently political and marked by moral and ethical dilemmas, alongside conflicting values and interests (Bærøe and Baltussen 2014; Baltussen, Jansen, Mikkelsen, et al. 2016). Funding decisions of novel, high-cost medicines exemplify the societal tensions involved, often surfacing as debates in the media (Abelson and Collins 2009; Corbett and Mori 1999). In this article, we provide experimental evidence on how citizens' support for public funding of novel cancer medicine varies depending on how the decision is framed. Previous research suggests that societal preferences regarding high-cost medications are susceptible to emotionally charged narratives circulated through mass media, which can, in turn, shape political decision-making (Gabe et al. 2012; Aggarwal et al. 2017; Pinho and Dias Costa 2020).

We show that preferences over healthcare funding decisions remain elastic to framing effects even when these decisions are insulated from politics and are not presented through individual patient stories. In democracies with publicly funded healthcare systems, politics are the arena for debating values that guide prioritisation and rationing. Nevertheless, reimbursement decisions are typically delegated to specialised arm's-length agencies that employ applied methods, such as health technology assessment (HTA), to inform decision-making (Egeberg and Trondal 2009; EC European Commission, Directorate-General for Health and Food Safety and Chamova 2018). These institutional arrangements are envisaged to depoliticise decision-making (Thatcher 2002). Yet, technical complexity and confidentiality can also distance the processes from the qualities associated with legitimacy (Barrenho and Lopert 2022; Mitton et al. 2006; Kapiriri and Razavi 2017; Daniels and Sabin 2008; Daniels and Wilt 2016).

Building from prospect theory, we argue that institutional insulation does not neutralise framing effects in mass preferences because it does not alter the cognitive mechanisms through which citizens evaluate distributive trade-offs. Instead, delegation shifts political contestation toward the realm of public interpretation. When citizens encounter decisions justified on technical or clinical grounds, the interpretive frame sets the reference point against which outcomes are judged—in this case, shaping whether distributive restraint is understood as prudent stewardship or as an avoidable loss.

Our findings have important implications for policy. Decisions on funding cancer medicines are increasingly challenged by the lack of robust clinical evidence, a consequence of policies that prioritise accelerated access of novel therapies (Hwang et al. 2020; Vella Bonanno, Ermisch, Godman, et al. 2017; Cohen 2017; Davis et al. 2017; Brinkhuis et al. 2024). Public and third-party payers must therefore decide on coverage despite limited data on health gains and safety, yet with full certainty of extraordinary costs (Cohen 2017). Although citizens and patients may not participate directly in these decisions, their perceptions of acceptability, fairness, and legitimacy influence political trust, willingness to pay taxes, confidence in the health system, and the uptake of supplementary private insurance as a response to perceived access failures.

Understanding the elasticity of mass preferences is essential for navigating how citizens interpret complex funding decisions. It can help public payers communicate more clearly about the reasoning behind difficult choices, while also providing insight into how different stakeholders, including commercial actors, advocacy groups, and policymakers, may use framing to advance their positions (Greer, Wismar, and Figueras 2016; Pinho and Dias Costa 2020).

In this study, we examine whether and how the framing of reimbursement decisions for novel, high-cost cancer medicines shapes citizens' support for public funding under conditions of technocratic

delegation, where decisions are formally insulated from partisan politics and justified through expert, procedural rationales rather than individual patient narratives. Using a survey experiment embedded in the 2021 Finnish Medicines Barometer, we randomly vary the interpretive frame applied to an otherwise identical funding decision and estimate the extent to which frames shift mass preferences by moving the reference point through which distributive trade-offs are evaluated. The core aim is to test the proposition, grounded in prospect theory, that institutional insulation does not neutralize framing effects because it does not change the underlying cognitive mechanisms of loss and gain evaluation, and thus that public interpretation remains a consequential arena in which the legitimacy and acceptability of rationing decisions are constructed.

Why this is important (legitimacy, trust, policy feedback) (1-2 paragraphs) MIKKO

This paper proceeds as follows. In section 2 we review the core principles and contested value trade-offs that structure healthcare priority setting and, in turn, supply the normative content for competing justifications of rationing decisions. In section 3 we develop the theoretical argument, specifying how prospect theory and the rule of rescue generate reference-point-dependent evaluations of high-cost reimbursement decisions under technocratic delegation. Next, section 4 describes the survey experiment embedded in the 2021 Finnish Medicines Barometer, including the framing manipulation and the empirical strategy used to estimate framing effects (Equation 1). We then present the results, reporting covariate balance (Figure 4) and model-based predicted probabilities across experimental conditions (Figure 2). In section 5 we offer some implications for the legitimacy and contestability of expert-justified rationing decisions in publicly funded healthcare systems.

2 Values underpinning priority setting in healthcare

Rationing inevitably involves value judgements, which may be explicit or implicit, either stated directly or embedded within assessment procedures (Clark and Weale 2012). For example, HTA frequently includes cost-effectiveness or cost-utility analysis as an evaluation method. The choice of such methods already reflects a normative commitment to health maximisation as the guiding prioritisation principle (Jansen et al. 2017; Daniels and Wilt 2016).

Prioritisation principles refer to judgements that reflect societal values, that is, moral or ethical values of the society in question (Clark and Weale 2012). For prioritisation to be legitimate, it must embody the values held by the population affected. Both the processes through which rationing decisions are made and the content of those decisions involve value judgements. The principles applied—whether economic, clinical, or related to personal characteristics, deservingness, or merit—remain widely contested and context dependent, making it difficult, if not impossible, to identify a single, overarching rule that would be accepted by all stakeholders (Oliver 2009; Watters 2016).

The overall value base of the European health systems, and the welfare states they are embedded, is grounded in solidarity and institutionalised as charters and legislation that ensure social rights. In healthcare, access, equity, quality, performance and efficiency have been proposed as the core principles through which European health systems pursue a fair distribution of resources (European Commission Expert Panel on Effective Ways of Investing in Health (EXPH) 2019). Values linked with legitimate decision-making processes include, for example, publicity, relevance, revisability, and enforceability (Daniels 2000; Daniels and Wilt 2016).

The focus of this article is, however, on values related to the content of decision-making (what decisions are made and why). Commonly applied values in healthcare prioritisation include clinical effectiveness and cost effectiveness (Clark and Weale 2012). These aim to direct scarce resources to interventions most likely to generate health gains, thereby producing the greatest possible

health for the population (health maximisation). The concept of opportunity cost refers to the foregone benefits associated with the next-best alternative, implying that resources invested in one intervention necessarily reduce what can be achieved elsewhere (Maguire, Persson, and Tinghög 2023). The application of these criteria depends on the availability of high quality evidence of clinical value. Furthermore, cost effectiveness is typically assessed through complex and technical evaluations of health benefits (e.g., Quality Adjusted Life Years, QALYs) and costs, with the aim of comparing the relative worth of different interventions. Although widely used, QALY concept involves controversial value judgements (Dukhanin et al. 2018; Mahdiani, Münch, and Paul 2024; Andrade 2025).

Social justice and equity considerations include the concepts of horizontal and vertical equity (O'Donnell et al. 2008). In healthcare, this implies that people with similar conditions, or conditions of similar severity, should be treated in the same way, while people with differing levels of disease severity should receive differential treatment according to their respective needs. Controversy arises when determining which personal characteristics are relevant to assessing those needs (Clark and Weale 2012). Are people of different ages, family structures, societal contributions, life expectancies, or lifestyle choices treated differently, or are such characteristics deemed irrelevant? For instance, the 'fair innings' intuition reflects preferences that view older people as already having enjoyed more life years than younger individuals, thereby prioritising the young so they too may be given the opportunity for a similar lifespan (Oliver 2009; Watters 2016; Adler et al. 2025).

Health maximisation criterion also clash with peoples' intuitive 'rule of rescue' obligation to save individuals from imminent death at any cost, even when those costs could be used more efficiently elsewhere (Cookson, McCabe, and Tsuchiya 2008; S. Sinclair 2022; Andrade 2025). The 'rule of rescue' has been attributed to preferences for identifiable patients, although it appears to apply more broadly (J. McKie and J. Richardson 2003; S. Sinclair 2022). One explanation centers on narrative considerations: people may be averse to let others to die suddenly without the opportunity settle their affairs, say goodbye, and come to terms with death (S. Sinclair 2022).

Payers have attempted to incorporate social justice and 'rule of rescue' considerations into the assessment criteria through various methods (Cookson, McCabe, and Tsuchiya 2008; Dukhanin et al. 2018; Mahdiani, Münch, and Paul 2024). Proportional or absolute shortfall methods seek to account for disease severity (Mahdiani, Münch, and Paul 2024) while end-of-life premiums allow relaxed cost-per-QALY thresholds for treatments expected to extend the lives of patients close to death (Olofsson, Gerdtham, Hultkrantz, et al. 2018). Nevertheless, even relaxed criteria require some evidence of added months, weeks or even days of life. Yet from the perspective of a severely ill patient and their family, it may be profoundly important that something, anything, is done, regardless of high levels of uncertainty about whether the intervention is beneficial or even harmful (Clark and Weale 2012; MacLeod, Harris, and Mahal 2016). Consequently, patients and citizens seem far less willing to accept any prioritisation based on cost than payers (MacLeod, Harris, and Mahal 2016). Furthermore, application of 'rule of rescue' as funding criteria has created concerns over the 'fetishisation of technology,' meaning the prioritising of making technologies available to all human conditions at the expense of population health (Cookson, McCabe, and Tsuchiya 2008).

Mass preferences related to public funding of cancer medicines are also likely influenced by peoples' perceptions of cancer. Intuitively, people tend to associate cancer with fear of imminent death, although this is often followed by a more deliberate recognition of improvements in care and increasing survival rates (Robb, Simon, Miles, et al. 2014). Willingness to pay, and reluctance to reduce spending, tends to be high for cancer care compared with many other conditions (Morrell, Wordsworth, Rees, et al. 2017; Schomerus, Matschinger, and Angermeyer 2006). However, this may reflect priority based on severity intuitively attached to cancer rather than specific preference for cancer itself, as evidence does not consistently support higher preference for cancer over other

conditions of similar severity (Morrell, Wordsworth, Rees, et al. 2017).

Empiric studies on public values suggest that societal willingness to pay decreases with patients' age and increases with perceived disease severity (Gu, Lancsar, et al. 2015; Loupas, Athanasakis, and Zavras 2026). Willingness to pay is high for the treatment of rare diseases despite high medical costs, although, there is no evidence of a clear preference for rare over common diseases (Gu, Wang, et al. 2023). Lower priority also appears to be given to people with self-induced conditions or higher socioeconomic positions (Gu, Lancsar, et al. 2015; Baute and Bellani 2025). The public tends to view individuals engaging in risky behaviours as less deserving of publicly financed healthcare and values preventive services for lifestyle-related conditions less than those targeting genetic disorders (Singh et al. 2012). Larger health gains tend to be prioritised over smaller gains, and life extensions over quality-of-life improvement, although evidence regarding end-of-life premiums is mixed, possibly due to sensitivity to framing effects (Gu, Lancsar, et al. 2015; Morrell, Wordsworth, Rees, et al. 2017). Opportunity costs are often neglected in personal consumption decisions, yet experimental evidence suggests that making them more salient exerts a small but robust effect on choice (Maguire, Persson, and Tinghög 2023).

While these studies help to illuminate public values, they are prone to bias, and it remains debatable whether, and to what extent, they should guide the selection of principles for priority setting (J. A. Whitty, Lancsar, Rixon, et al. 2014; Gustavsson and Lindblom 2025). This is particularly relevant when mass preferences conflict with human and/or social rights (e.g. racist preferences). Deservingness opinions are also shaped by individuals' level of understanding of the topic, which may explain why the public and policymakers often prioritise different values. Citizens' views may be more influenced by simple deservingness heuristics, while policymakers more familiar with the issues tend to consider a broader array of perspectives in their decisions (Shah et al. 2015; Van Der Aa et al. 2017; Jensen and Petersen 2017).

Intuitive needs rooted in fear of death and the desire to preserve hope also create a vulnerability that can be commercially exploited. In the pharmaceutical sector, one factor to acknowledge is the intervention of well-resourced industry lobbies, capable of deploying costly strategies to pursue and protect their interests (Greer, Wismar, and Figueras 2016). Close ties between pharmaceutical industry and patient organisations are well known, including likely commercially motivated funding pursuits (Mulinari et al. 2020; Fabbri et al. 2020). These stakeholders share interests in terms of promoting patient access to novel therapies, however, this also creates pathways through which the industry may exert pressure for public funding of new medicines with uncertain value. A more recent concern involves promotion through social media, for example by sponsoring patient influencers (Mor et al. 2024).

Novel cancer medicines entering markets with high uncertainty about effectiveness and safety cannot be meaningfully assessed in terms of cost-effectiveness. Yet stories of dying patients denied 'last-chance' interventions tend to resonate strongly with the public, shaping societal expectations about what healthcare ought to provide, regardless of cost or evidence (Pinho and Dias Costa 2020; Rubagumya et al. 2023). In these debates, public payers are often portrayed as 'the villains,' preventing patients from reaching the newest innovations that could rescue them from imminent death (Cohen 2017).

These tensions highlight the need to better understand the mass preferences concerning values that underpin healthcare prioritisation, and how these preferences may be intentionally or unintentionally exploited in favour of commercial stakeholders over societal interests. If healthcare continues to be organised through solidarity-based systems, the challenge lies in determining how such systems can uphold their core values and maintain legitimacy in a context of decreasing public resources, rapidly expanding technological possibilities, rising public expectations and strengthening commercial power.

The experiences of the NHS Cancer Drugs Fund in England provide an illustrative example.

The Fund was created in response to public and political pressure to finance new cancer medicines that National Institute of Health and Care Excellence (NICE) had not recommended on the basis of its assessment criteria. An evaluation conducted five years after the establishment of Fund found no meaningful benefit to patients or society, when measured in terms of survival, quality of life, or toxicity, and suggested potential societal harm once opportunity costs were taken into account (Aggarwal et al. 2017). Thus, while new medications may have brought hope to patients, this hope appeared to a large extent unfounded. Overall, retrospective analyses of novel cancer medicines approved through accelerated access pathways suggest very limited added clinical benefits (Davis et al. 2017; Hwang et al. 2020; Brinkhuis et al. 2024).

3 Theoretical framework: prospect theory and the rule of rescue in delegated healthcare

Prospect theory provides the micro-foundations for understanding how citizens evaluate distributive choices in healthcare rationing and why such evaluations are reference-point dependent. Individuals do not assess outcomes in terms of final states, but as gains and losses relative to a reference point (Kahneman and Tversky 1979, p. 263). The canonical prospect-theoretic value function is concave for gains, convex for losses, and steeper in the loss domain (Kahneman and Tversky 1984, p. 342), implying diminishing sensitivity and, crucially, loss aversion: losses loom larger than equivalent gains. These regularities remain among the most robust findings in decision research (Barberis 2013). In the context of healthcare rationing, this implies that the same reimbursement decision can be psychologically represented either as avoiding a loss (not withholding a last-resort option) or as protecting a gain (preserving finite collective resources), and these alternative representations can generate systematically different preferences even when clinical information and costs are held constant.

Reference points are not fixed. They are shaped by expectations and by communicated baselines (Koszegi and Rabin 2006; Koszegi and Rabin 2007). In policy environments, where most citizens encounter specialized decisions through mediated communication, the descriptive frame becomes part of how the baseline is constructed: it indicates what is treated as the default and what is treated as a deviation (Kahneman 2003). Applied to delegated healthcare rationing, citizens may anchor on multiple plausible baselines: existing coverage norms, a generalized expectation of equal access, an identifiable patient in need, or the collective budget constraint emphasized by expert institutions. The same choice can therefore be interpreted as a departure from an entitlement-like status quo (experienced as a loss) or as a refusal to extend coverage beyond what is fiscally sustainable (experienced as a foregone gain).

Prospect theory also implies a domain-specific pattern of risk-taking. Because the value function is concave over gains and convex over losses, individuals tend to be risk-averse in the gain domain and risk-seeking in the loss domain (the reflection effect) (Kahneman and Tversky 1979; Vis 2011). This matters directly for high-cost oncology medicines, where decisions often involve clinical uncertainty and the possibility of small-probability benefits at high financial cost. When the situation is framed in the loss domain (for example, as denying a last-resort option), citizens should be more willing to tolerate fiscal risk and uncertainty to avert the salient loss (Hameleers 2021). When the same situation is framed in the gain domain (for example, as protecting finite resources from extraordinary expenditure and preserving broader access), citizens should become more supportive of restraint (Hameleers 2021). Related mechanisms reinforce this asymmetry. Probability weighting implies that small probabilities can receive disproportionate attention (Kahneman and Tversky 1979; Kahneman and Tversky 1984); in a rescue-oriented loss frame, even a modest chance of benefit can become

politically and morally salient relative to diffuse opportunity costs that remain abstract.

These mechanisms should persist under technocratic delegation because frames operate through interpretation rather than through formal institutional responsibility. Delegation is often narrated as depoliticization: evidence is assessed through objective criteria, thresholds are applied consistently, and distributive choices are presented as technically grounded rather than politically discretionary (Thatcher and Sweet 2002). Yet the fact that decisions are justified technically does not immunize publics against reference dependence. Technical criteria themselves must be interpreted, and citizens draw pragmatic inferences about what a communicated rationale implies and what the decision-maker is prioritizing (Flusberg et al. 2024). Moreover, complexity increases reliance on accessible cues. Under uncertainty, citizens frequently lean on heuristics such as availability, which gives greater weight to vivid, easily imagined cases than to abstract aggregates (Tversky and Kahneman 1973). In rationing, the narrative of a specific patient denied a last-resort medicine is cognitively accessible, whereas opportunity costs borne by anonymous others are comparatively opaque. This asymmetry makes it easier for loss frames to dominate public evaluation even when expert institutions supply cost-effectiveness language and probabilistic evidence.

If prospect theory explains why equivalent choices can elicit different preferences depending on the reference point, the rule of rescue explains why certain reference points are uniquely powerful in healthcare. The rule of rescue is the perceived imperative to save identifiable individuals facing avoidable death even when doing so is inefficient and diverts resources from saving more lives elsewhere (John McKie and Jeff Richardson 2003, p. 2407). It entails a preference for identifiable over statistical lives and a "shock-horror" response that prioritizes lifesaving over non-lifesaving measures (John McKie and Jeff Richardson 2003, p. 2407). In rationing settings, rescue narratives convert a distributive trade-off into an evaluation of whether the state is allowing an avoidable tragedy to occur. Because rescue frames foreground an identifiable beneficiary and a morally charged counterfactual (a life that could have been saved), they are well suited to placing citizens in the loss domain, thereby activating loss aversion and risk-seeking responses predicted by prospect theory. Extensions also suggest that end-of-life contexts can intensify the perceived moral stakes of denial, including concerns about narrative closure (Sean Sinclair 2022), which further increases the salience of losses relative to diffuse population-level gains.

Taken together, prospect theory and the rule of rescue provide a coherent account of why technocratic delegation should not eliminate framing effects in public evaluations of reimbursement decisions. When an identical reimbursement decision is framed as avoiding a salient loss to identifiable patients, citizens should be more willing to endorse riskier and more expensive options. When it is framed as preserving collective gains by protecting finite resources, citizens should be more willing to accept restraint. Rescue narratives should magnify the effect of loss framing by intensifying moral salience and by making the focal outcome vivid, concrete, and attributable. These mechanisms operate at the level of cognition and interpretation, and thus remain active even when decisions are formally insulated from day-to-day electoral accountability and justified through expert procedures. In that sense, technocratic delegation can redistribute political pressure without removing contestation: the struggle shifts toward how decisions are communicated, what reference points are made salient, and whether expert-justified restraint is interpreted as legitimate stewardship or as a failure to rescue.

This implication aligns with research on delegation and accountability, which suggests that delegation may facilitate blame avoidance only under specific institutional conditions and only insofar as citizens accept the attributional story that responsibility has shifted (Heinkelmann-Wild et al. 2023; Heinkelmann-Wild and Zangl 2020). It also aligns with work on the contestability of expertise, where public acceptance of expert advice depends on more than technical merit and remains vulnerable to competing narratives (Head 2024). Accordingly, even in delegated settings, mass

preferences over high-cost medicines should remain elastic to loss-versus-gain frames (Laenen 2025), and frames that activate rescue and loss aversion should systematically shift support toward more permissive funding positions, while stewardship frames should shift support toward conditionality and rejection.

The theoretical framework developed above assumes that formal institutional insulation does not eliminate political contestation; it redirects it toward the arena of public interpretation. This claim benefits from engagement with broader institutional theory on delegation and accountability. Delegation to technically specialized agencies is a common governance strategy in contemporary democracies, reflecting functional demands for expertise as well as political incentives to shift responsibility for contentious decisions. Yet political scientists have questioned whether institutional insulation genuinely depoliticizes decision-making or instead obscures underlying political struggle (Koop and Hanretty 2018). Comparative work suggests that formal independence can improve regulatory quality by enabling more technically sophisticated decisions (Koop and Hanretty 2018). At the same time, such capacity gains do not remove the distributive and value-laden character of decisions with major consequences for affected publics (Langvatn and Holst 2022), and insulation can generate accountability deficits by weakening electoral responsiveness and limiting conventional channels of democratic control (Crum 2017). In healthcare, where rationing must reconcile efficiency criteria with legitimacy concerns, these institutional tensions reinforce the central implication of our argument: framing effects persist under technocratic delegation because they operate through reference-dependent cognitive evaluation rather than through formal responsibility alone.

4 Experimental evidence on framing and mass preferences in delegated healthcare decisions

The empirical analysis draws on data from the 2021 Finnish Medicines Barometer, a national, cross-sectional population survey administered biennially by the Finnish Medicines Agency to examine experiences, opinions, and values related to health, medicines, and well-being. The 2021 wave included an ad hoc experimental module designed specifically to assess how citizens evaluate expert-justified healthcare funding decisions when identical cases are presented under alternative frames. This design directly reflects the theoretical framework developed above: in contemporary healthcare systems, reimbursement decisions are typically made through expert procedures and justified using technical criteria, yet they remain visible to the public and subject to mass evaluation. The survey questionnaire was drafted with influence from previous studies conducted in other countries, in particular the survey by Blendon et al. (2012), which similarly exposed citizens to real-life decision-making tasks, and Ghinea et al. (2021), which surveyed Australian citizens' attitudes towards public funding of high-cost cancer medicines.

Finland provides an appropriate research context for three reasons. First, the healthcare system is comprehensive and predominantly publicly funded, resembling other advanced welfare states in which questions of healthcare rationing and legitimacy are politically salient. Second, healthcare resource allocation decisions are routinely justified through clinical evidence, cost-effectiveness thresholds, and health technology assessment, making the communicative dimension of these decisions especially important. However, despite the normative emphasis on transparency as a core dimension of legitimate limit-setting, decision-making processes for pharmaceutical reimbursement in Finland have been described as lacking full transparency in how evidence and stakeholder input are considered and used, including in official appraisal and pricing forums. Third, Nordic healthcare systems face precisely the structural pressures that motivate this study: technological and demographic changes contribute to a widening gap between medical opportunities and fiscal possibilities, which is a

common source of controversy among stakeholders.

Data collection was carried out by a professional market research company (Taloustutkimus Ltd) using a pre-recruited online panel of approximately 40,000 Finnish citizens. To reach the target sample size, 10,105 invitations were distributed to panel members. The sample was stratified to ensure balance across gender, age, education level, and geographic region, with eligibility restricted to Finnish-speaking citizens aged 18—79 years. The final analytic sample consists of 2,081 respondents, corresponding to a completion rate of approximately 20.6 percent. While online panels are not probability samples, stratification and quota-based recruitment ensure close correspondence with the Finnish adult population on key sociodemographic dimensions, making the data suitable for inference about mass public opinion in a welfare-state context.

The experimental module employed a randomized between-subjects design focused on framing effects. Respondents were randomly assigned to one of three conditions: a control condition and two treatment conditions that varied only in how the decision context was framed. All respondents received an identical clinical vignette describing a new oncology medicine characterized by high cost, uncertain benefit, and limited patient eligibility (Table 1). The vignette was drafted based on a published assessment by the Council for Choices in Health Care in Finland (COHERE Finland), with similar decision options that are used by their recommendations. COHERE, working in conjunction with the Ministry of Social Affairs and Health, only issues recommendations on including or excluding health technologies in the range of public health services. Decisions on reimbursement for outpatient medicines are conducted by the Pharmaceutical Pricing Board, also working in conjunction with the Ministry of Social Affairs and Health, and for inpatient medicines, the ultimate decision lies with hospitals. The framing manipulation consisted solely of additional information presented between the vignette and the decision task, allowing differences in responses to be attributed to framing rather than to substantive informational differences.

Following the vignette and framing manipulation, respondents were asked which decision regarding the medicine's use they would find acceptable. Response categories captured increasing levels of cost containment: unconditional public funding, conditional funding tied to price reductions, rejection of funding, and an explicit "I don't know" option. These categories map directly onto the theoretical expectations derived from prospect theory, as unconditional funding represents the most risk-acceptant position, conditional funding an intermediate position, and rejection the most risk-averse position.

To estimate the effect of framing on policy preferences, we model respondents' choices using an ordinal logistic regression, reflecting the ordered structure of the outcome variable (Equation 1). The key independent variable is the framing condition, operationalized as indicators for the rule-of-rescue frame and the stewardship (utility-maximizing) frame, with the control condition as the reference category. The specification also includes demographic and socioeconomic covariates commonly associated with healthcare attitudes: gender, age, and income group, as well as indicators capturing health-related exposure, including eligibility for reimbursement of medicines and self-reported medicine expenditures. All models are estimated with survey weights to preserve population representativeness.

Formally, the estimated model can be written as:

$$\Pr(Y_i \leq j) = \text{logit}^{-1}(\tau_j - \beta_1 \,\text{LossRescue}_i - \beta_2 \,\text{GainsStewardship}_i - \boldsymbol{\gamma}\mathbf{X}_i)\,, \tag{1}$$

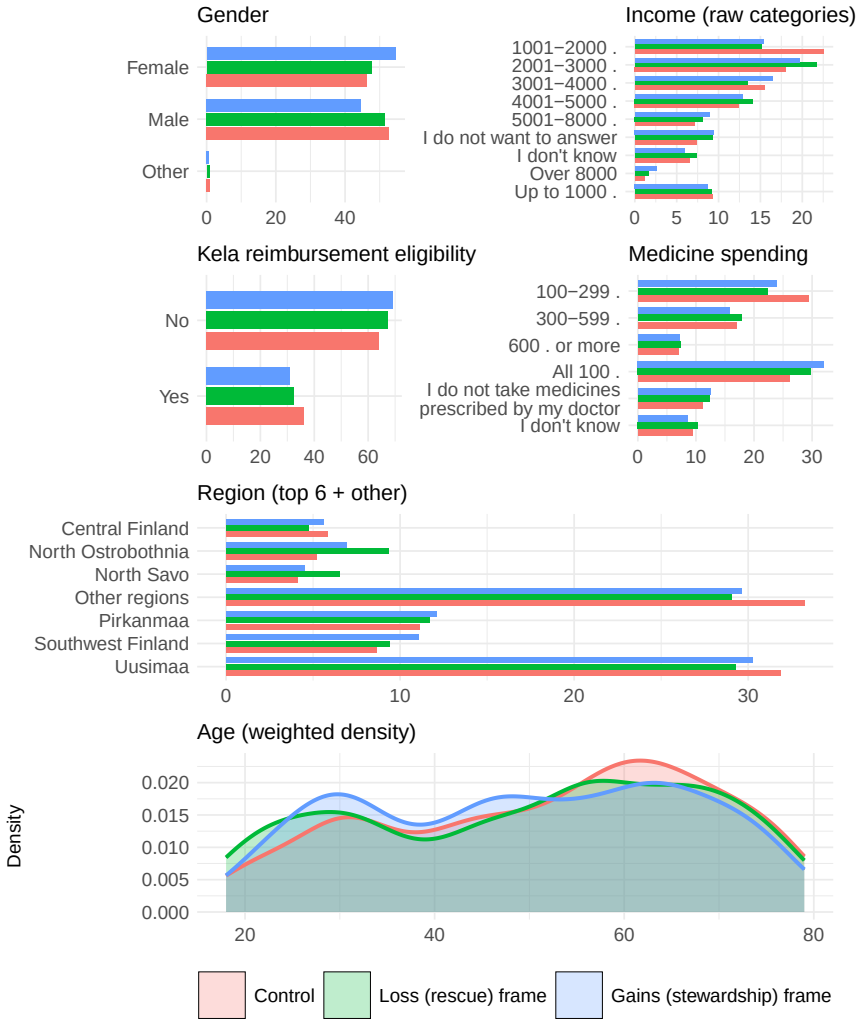
where Y_i denotes respondent i's preferred funding decision, ordered from unconditional funding to rejection; τ_j represents threshold parameters; LossRescue_i and $\text{GainsStewardship}_i$ indicate assignment to the loss-framed and gain-framed conditions, respectively; and $\mathbf{X}_i$ is a vector of individual-level covariates including gender, age, income group, health status indicators, and

Table 1: Decision-making task administered in the survey with framing manipulation.

	Group A (rule-of-rescue frame)	**Group B (health maximisation frame)**	**Group C (control)**
Vignette	Below, you will find a description of the benefits and costs of a new cancer medicine. Please read the description and then answer the question. There are annually 10 to 20 patients in Finland to whom the new medicine can be prescribed. The medicine does not cure cancer but a laboratory study has shown that it destroys cancer cells in about a third of the patients to whom the new medicine is given. It remains unknown if the new medicine extends patients' lives or if it improves their quality of life. The medicine comes with a great risk of adverse effects. If this new medicine is to be approved for use in Finland, its costs will be over 60.000 euros per patient at the final stage of cancer treatment.		
Frame	There is no cure for this particular type of cancer. The new medicine is a possible option for patients who have already received multiple treatments and for whom the remaining options are limited.	The funds available to healthcare are finite. The adoption of the new medicine means that the funds used to pay for it will mean cuts elsewhere in healthcare.	None
Task	What kind of decision regarding the new medicine's use would you find acceptable? Please choose the option that best reflects your opinion. 1. The medicine should be adopted to use and paid by the society regardless of the price set on it by the pharmaceutical company 2. The medicine should be adopted to use and paid by the society if the pharmaceutical company lowers its price 3. The medicine should not be adopted to use and paid by the society 4. I don't know		

medicine expenditures. Equation 1 therefore estimates how framing shifts the latent propensity to favor more restrictive funding positions, holding observed characteristics constant.

Figure 1: Frequencies and Densities: Summary Statistics and Covariates Balance

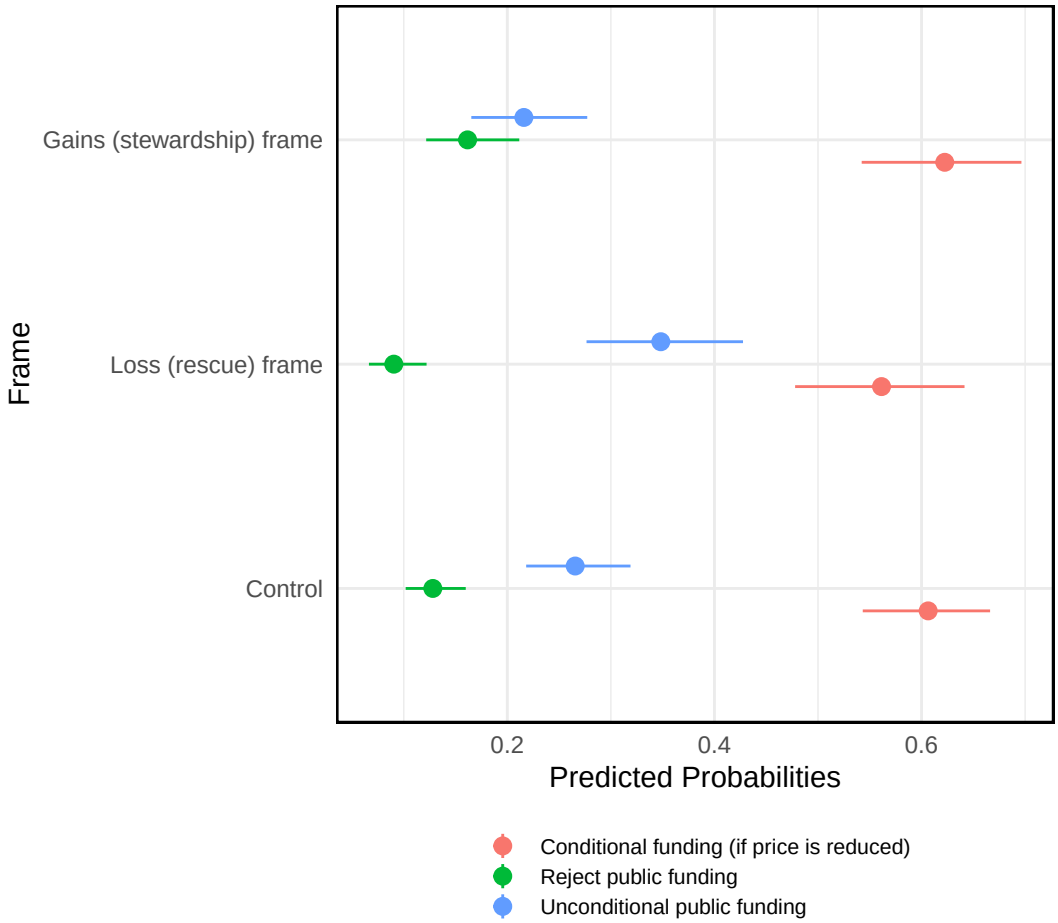


Notes: The figure summarizes sample composition by experimental condition (Control, Loss/rescue, and Gains/stewardship). The age panel displays weighted kernel density estimates of respondents' age. The remaining panels report weighted percentages within each framing condition for the covariates used in the main model (Equation 1). All estimates use survey weights. The close overlap across conditions suggests good covariate balance from random assignment.

4.1 Results

After fitting Equation 1, we show Figure 2 which presents predicted probabilities for the three principal policy preferences across framing conditions (regression table shown in Table A1). The results reveal clear and systematic framing effects. Holding the clinical vignette, cost information, and decision task constant, respondents exposed to alternative frames shifted their preferred policy responses in substantively meaningful ways.

Figure 2: Framing Effects on Public Support for Funding a High-Cost Cancer Medicine



Notes: Points are predicted probabilities from the ordered logistic model (Equation 1; regression table shown in Table A1); vertical bars indicate 90% confidence intervals. The loss (rescue) frame increases support for unconditional funding and reduces outright rejection relative to the control, whereas the gains (stewardship) frame shifts support toward conditional funding and rejection.

The most pronounced pattern concerns support for unconditional public funding, the most risk-seeking option in the decision space. Consistent with the expectations derived from prospect theory, support for unconditional funding increases under the loss frame that emphasizes the absence of alternatives and the proximity of death, and decreases under the gains frame that emphasizes finite budgets and opportunity costs. Conversely, rejection of funding—the most risk-averse response—becomes more likely under the gains frame and less likely under the loss frame. Conditional funding occupies an intermediate position, absorbing shifts in both directions as respondents adjust their evaluations of fiscal risk.

Across conditions, the rank ordering of response options remains stable: unconditional funding is consistently most likely, followed by conditional funding, with outright rejection least likely. What changes is not the rank itself but the level of support attached to each option, as the frames sift probability mass across categories. In particular, the loss frame sifts support upward toward unconditional funding, whereas the gains frame sifts support away from unconditional funding and toward conditional funding and rejection. Thus, framing shifts the level of support across options

while preserving the same ordinal structure of preferences.

These results suggest that citizens do not evaluate reimbursement decisions solely in terms of technical information about clinical effectiveness or cost. Instead, preferences vary systematically with the reference point activated by communication (frames). When the decision is presented as avoiding a salient loss to identifiable patients, respondents appear more willing to accept financial risk and clinical uncertainty. When the same decision is presented as protecting existing resources for the broader patient population, respondents become more cautious and more supportive of restraint. The observed shifts are therefore consistent with the reflection effect and loss aversion predicted by prospect theory, as well as with the moral psychology captured by the rule of rescue.

Importantly, these effects emerge in a setting where respondents are not asked to act as policymakers but to evaluate a decision justified through information resembling real reimbursement debates. This feature of the design speaks directly to the institutional argument advanced in this article. The findings suggest that insulating reimbursement decisions from day-to-day electoral politics and grounding them in technical criteria does not eliminate the role of framing in shaping public evaluation. Citizens remain responsive to how choices are narrated, and their preferences display substantial elasticity even when the informational content of the decision remains constant.

We believe this elasticity has important implications for the legitimacy and contestability of healthcare rationing. Delegation to expert bodies may reduce direct political responsibility for specific reimbursement decisions, but it does not remove those decisions from the arena of public judgment. Instead, it shifts the locus of contestation toward communication, interpretation, and narrative framing. Public reactions to expert-justified decisions depend not only on the underlying evidence but also on whether the decision is understood as prudent stewardship of collective resources or as a morally troubling failure to rescue.

Taken together, the results provide empirical support for the theoretical claim developed in the previous section: reference-point–dependent evaluation and rescue-oriented reasoning persist even under institutional arrangements designed to depoliticize distributive choices. In contexts characterized by uncertainty, high stakes, and competing moral logics, framing remains a central mechanism through which citizens interpret and evaluate state action.

5 Discussion

This article examined whether technocratic delegation constrains framing effects in mass evaluations of costly reimbursement decisions, or whether citizens' preferences remain reference-point dependent even when authority is insulated from day-to-day electoral politics. Using a population-based survey experiment embedded in the 2021 Finnish Medicines Barometer, we held constant the clinical vignette, uncertainty, and cost information while varying only the interpretive frame attached to an otherwise identical decision. The results indicate substantial and systematic framing effects. Relative to the control condition, the loss (rule-of-rescue) frame shifts respondents toward unconditional public funding and away from outright rejection, whereas the gains (stewardship) frame increases support for conditionality and rejection. Randomization performed as intended, and model-based predicted probabilities provide a substantively interpretable account of how frames redistribute support across policy options.

These findings support the theoretical claim developed in section 3: technocratic delegation does not neutralize framing effects because framing operates through cognitive evaluation and public interpretation rather than through formal responsibility. Even when decisions are presented in technical language and without identifiable patient narratives, citizens' judgments remain elastic to whether the same distributive trade-off is framed as avoiding losses to patients with limited options

or as preserving collective gains by protecting finite resources. Contestation therefore lies not only in institutional design (who decides) but also in interpretive processes (how decisions are explained and which reference points are made salient). In delegated domains, legitimacy becomes partly communicative: publics infer priorities from how expert rationales are framed, and these inferences shift preferences even when underlying evidence remains constant.

The implications extend to democratic theory. Tensions between expertise and popular sovereignty have long been recognized (Gaus, Landwehr, and Schmalz-Bruns 2020). Delegation is often defended as a means of depoliticizing contentious distributive decisions by placing them in the hands of insulated experts. Yet an alternative perspective suggests that insulation does not remove politics but relocates it from formal arenas of electoral accountability to informal arenas of interpretation. Our results support this latter view. Even when respondents receive information resembling expert assessments, modest variations in framing produce meaningful shifts in support. Institutional insulation therefore does not eliminate contestation; it displaces it to the communicative domain, where asymmetries in expertise and resources may advantage organized actors over dispersed publics (Langvatn and Holst 2022; Eriksen 2020). Sustaining legitimacy under delegation thus requires not only technical rigor but also transparency, participation, and mechanisms through which citizens can scrutinize and contest expert reasoning.

The findings also illuminate the politics of high-cost oncology medicines under evidentiary uncertainty. When benefits are uncertain and costs are salient, rescue-oriented frames increase willingness to accept fiscal and clinical risk, consistent with prospect-theoretic expectations about loss aversion and loss-domain risk seeking. By contrast, frames emphasizing opportunity costs and resource constraints increase support for restraint. Public reactions are therefore not simple reflections of stable distributive principles; they are partly constructed at the point of interpretation, where competing normative logics—rescue versus maximization—are activated. In practice, technocratic delegation may shift political pressure toward communication strategies and narrative competition among stakeholders rather than eliminating politics from the domain.

This dynamic intersects with scholarship on expert decision-making in contested policy areas. Governance research emphasizes the importance of "political literacy": recognition that technically justified decisions inevitably embed normative judgments (Eriksen 2020). Political literacy does not imply partisanship but awareness that regulation frames problems and relies on contestable interpretive concepts (Eriksen 2020). In healthcare, funding decisions necessarily reflect judgments about risk, fairness, and distributive priorities. Our findings indicate that citizens are responsive to how these trade-offs are presented. Expert agencies therefore cannot rely on claims of neutrality alone; legitimacy depends in part on explaining the value commitments implicit in technical reasoning. Communication is not peripheral to delegated authority but integral to how decisions are evaluated.

Moreover, real-world rationing debates rarely occur under neutral framing conditions. Competing narratives are advanced by organized interests with unequal communicative resources. The pharmaceutical industry, patient advocacy organizations, professional associations, and public authorities deploy frames strategically to shape interpretation. Industry lobbying exceeds $233 million annually in the United States, alongside political engagement and advocacy funding (Steinbrook 2020). Funded patient organizations frequently participate in coordinated campaigns amplifying rescue-oriented narratives (Lisa Parker et al. 2019), within broader patterns of industry support documented across contexts (Lisa Parker et al. 2019; Fabbri et al. 2020; Mulinari et al. 2020). Given the emotional salience of rescue frames in life-threatening conditions, such asymmetries can influence public evaluations of identical evidence. This does not imply that framing is inherently manipulative; rather, it underscores that delegated governance operates within competitive interpretive environments. Institutional safeguards may therefore need to address communicative balance in addition to procedural rigor.

One institutional response to persistent framing effects is to strengthen citizen participation in priority-setting itself. Rather than insulating decisions and subsequently managing public opinion, governance can involve citizens upstream in defining and applying rationing criteria. Hybrid approaches that combine deliberation with preference elicitation are promising. Citizens' juries have demonstrated the capacity to produce informed and nuanced judgments about healthcare trade-offs (Jennifer A. Whitty et al. 2014), and when paired with discrete choice experiments can link deliberation to quantified societal weights (Jennifer A. Whitty et al. 2014). However, systematic evidence indicates that meaningful participation requires genuine influence, adequate deliberative conditions, and accessible information (Arthur, Saha, and Kapilashrami 2023). In technically complex domains such as oncology funding, these conditions are demanding. Nonetheless, if framing effects persist under insulation, democratizing elements of priority-setting may provide a more durable legitimacy strategy than reliance on technocratic authority alone.

The extent to which these dynamics generalize beyond Finland remains an open question. Finland represents one configuration of publicly financed, expert-driven healthcare governance. Comparative research shows that institutional design—agency independence, transparency, accountability mechanisms, and public participation—shapes both contestation and legitimacy (Crum 2017; Koop and Hanretty 2018). Future research should examine whether similar framing elasticities arise in other welfare states and in systems organized around different governance logics, including more market-oriented or corporatist arrangements. Comparative analysis could clarify whether embedding earlier citizen involvement attenuates interpretive contestation relative to systems prioritizing insulation and post hoc communication.

Several limitations qualify these conclusions. First, the study relies on a single-country sample and a single policy domain, limiting external validity across institutional contexts and disease areas. Second, the experiment employs one vignette configuration; framing effects may vary with uncertainty type, budget impact, or perceived deservingness. Third, treatments are brief, which enhances internal validity but may not capture repeated exposure to emotionally saturated messaging or elite cues. Fourth, the outcome is attitudinal rather than behavioral, so downstream political consequences are inferred rather than observed. Finally, although the ordinal logit framework appropriately models ordered responses, psychological mediators such as perceived loss magnitude or moral obligation were not directly measured.

These limitations suggest a focused research agenda. Future studies should examine heterogeneity in framing sensitivity, given that reference points may vary with political trust, system satisfaction, illness exposure, redistribution attitudes, or policy knowledge. Attribution and accountability mechanisms should also be manipulated directly, varying who is said to decide and what procedural safeguards are emphasized. Additional work can explore interactions between framing and evidence communication, including how uncertainty and opportunity costs are described. Comparative experiments contrasting technocratic framing with patient-story framing would clarify whether delegation dampens or merely displaces the persuasive force of identifiable beneficiaries. Finally, longitudinal and field designs could assess persistence, learning, and real-world political consequences.

Overall, the evidence suggests that even under technocratic delegation, distributive healthcare decisions remain politically consequential because citizens evaluate them through reference-dependent interpretive frames. Delegation may reduce overt partisan conflict over particular reimbursement decisions, but it does not eliminate the cognitive and moral structures through which rationing is judged. Rather, it shifts contestation toward how expert justifications are framed and which reference points communication renders salient.

References

Abelson, J. and P. A. Collins (2009). “Media Hyping and the "Herceptin Access Story": An Analysis of Canadian and UK Newspaper Coverage.” In: *Healthc Policy.* 4.3. Accessed September 17, 2023, e113–e128.

Adler, M. D. et al. (2025). “Fair innings: an empirical test.” In: *Health economics* 34.7, pp. 1350–1364.

Aggarwal, A. et al. (2017). “Do patient access schemes for high-cost cancer drugs deliver value to society?—lessons from the NHS Cancer Drugs Fund.” In: *Ann Oncol.* 28.8, pp. 1738–1750.

Andrade, G. (2025). “Ethical Shortcomings of QALY: Discrimination Against Minorities in Public Health.” In: *Cambridge Quarterly of Healthcare Ethics* 34.1, pp. 109–116.

Arthur, Megan, Ria Saha, and Anuj Kapilashrami (2023). “Community participation and stakeholder engagement in determining health service coverage: A systematic review and framework synthesis to assess effectiveness.” In: *Journal of Global Health* 13, p. 04034.

Bærøe, K. and R. Baltussen (2014). “Legitimate healthcare limit setting in a real-world setting: integrating accountability for reasonableness and multi-criteria decision analysis.” In: *Public Health Ethics* 7.2, pp. 98–111.

Baltussen, R., M. P. Jansen, E. Mikkelsen, et al. (2016). “Priority setting for universal health coverage: we need evidence-informed deliberative processes, not just more evidence on cost-effectiveness.” In: *International journal of health policy and management* 5.11, p. 615.

Barberis, Nicholas (Feb. 2013). “Thirty Years of Prospect Theory in Economics: A Review and Assessment.” In: *Journal of Economic Perspectives* 27.1, pp. 173–196.

Barrenho, E. and R. Lopert (2022). *Exploring the Consequences of Greater Price Transparency on the Dynamics of Pharmaceutical Markets.* OECD, p. 45.

Baute, S. and L. Bellani (2025). “Healthcare deservingness: how risk factors and income shape responsibility attribution for health outcomes and healthcare costs.” In: *European Sociological Review* 41.3, pp. 457–471.

Blendon, Robert J. et al. (May 2012). “A four-country survey of public attitudes towards restricting healthcare costs by limiting the use of high-cost medical interventions.” In: *BMJ Open* 2.3, e001087.

Brinkhuis, F. et al. (2024). “Added benefit and revenues of oncology drugs approved by the European Medicines Agency between 1995 and 2020: retrospective cohort study.” In: *bmj* 384.

Clark, S. and A. Weale (2012). “Social values in health priority setting: a conceptual framework.” In: *Journal of health organization and management* 26.3, pp. 293–316.

Cohen, D. (2017). “Cancer drugs: high price, uncertain value.” In: *BMJ* 359, j4543.

Cookson, R., C. McCabe, and A. Tsuchiya (2008). “Public healthcare resource allocation and the Rule of Rescue.” In: *Journal of Medical Ethics* 34, pp. 540–544.

Corbett, J. B. and M. Mori (1999). “Medicine, Media, and Celebrities: News Coverage of Breast Cancer, 1960–1995.” In: *Journal Mass Commun Q.* 76.2, pp. 229–249.

Crum, Ben (2017). “Parliamentary accountability in multilevel governance: What role for parliaments in post-crisis EU economic governance?” In: *Journal of European Public Policy* 24.3, pp. 455–475.

Daniels, N. (2000). “Accountability for reasonableness.” In: *BMJ* 321.7272, pp. 1300–1.

Daniels, N. and J. E. Sabin (2008). “Accountability for reasonableness: an update.” In: *BMJ* 337, a1850.

Daniels, N. and G. J. van der Wilt (2016). “Health technology assessment, deliberative process, and ethically contested issues.” In: *International Journal of Technology Assessment in Health Care* 32.1-2, pp. 10–15.

Davis, C. et al. (2017). “Availability of evidence of benefits on overall survival and quality of life of cancer drugs approved by European Medicines Agency: retrospective cohort study of drug approvals 2009-13.” In: *BMJ* 359, j4530.

Dukhanin, V. et al. (2018). “Integrating social justice concerns into economic evaluation for healthcare and public health: A systematic review.” In: *Soc Sci Med.* 198, pp. 27–35.

EC European Commission, Directorate-General for Health and Food Safety and J. Chamova (2018). *Mapping of HTA National Organisations, Programmes and Processes in EU and Norway - Publications Office of the EU*. Publications Office.

Egeberg, M. and J. Trondal (2009). “Political Leadership and Bureaucratic Autonomy: Effects of Agencification.” In: *Governance* 22.4, pp. 673–688.

Eriksen, Andreas (2020). “The Political Literacy of Experts.” In: *Ratio Juris* 33.1, pp. 82–97.

European Commission Expert Panel on Effective Ways of Investing in Health (EXPH) (2019). *Defining value in "value-based healthcare"*. Publications Office of the European Union.

Fabbri, A. et al. (2020). “Industry funding of patient and health consumer organisations: systematic review with meta-analysis.” In: *BMJ* 368, p. l6925.

Flusberg, Stephen J. et al. (Dec. 2024). “The Psychology of Framing: How Everyday Language Shapes the Way We Think, Feel, and Act.” In: *Psychological Science in the Public Interest* 25.3, pp. 105–161.

Gabe, J. et al. (2012). “The debate about the funding of Herceptin: a case study of "countervailing powers".” In: *Soc Sci Med.* 75.12, pp. 2353–2361.

Gaus, Daniel, Claudia Landwehr, and Rainer Schmalz-Bruns (2020). “Defending democracy against technocracy and populism: Deliberative democracy’s strengths and challenges.” In: *Constellations* 27.3, pp. 335–347.

Ghinea, Narcyz et al. (Mar. 2021). “A survey of Australian public attitudes towards funding of high cost cancer medicines.” In: *Health Policy* 125.3, pp. 327–334.

Greer, S., M. Wismar, and J. Figueras (2016). *Strengthening Health System Governance: Better Policies, Strong-er Performance*. World Health Organization, European Observatory on Health Systems and Poli-cies.

Gu, Y., E. Lancsar, et al. (2015). “Attributes and weights in health care priority setting: a systematic review of what counts and to what extent.” In: *Social Science & Medicine* 146, pp. 41–52.

Gu, Y., A. Wang, et al. (2023). “Comparison of Rare and Common Diseases in the Setting of Healthcare Priorities: Evidence of Social Preferences Based on a Systematic Review.” In: *Patient Preference and Adherence* 17, pp. 1783–1797.

Gustavsson, E. and L. Lindblom (2025). “Justification of principles for healthcare priority setting: the relevance and roles of empirical studies exploring public values.” In: *Journal of medical ethics* 51.4, pp. 285–292.

Hameleers, Michael (July 2021). “Prospect Theory in Times of a Pandemic: The Effects of Gain versus Loss Framing on Risky Choices and Emotional Responses during the 2020 Coronavirus Outbreak – Evidence from the US and the Netherlands.” In: *Mass Communication and Society* 24.4, pp. 479–499.

Head, Brian W. (June 2024). “Reconsidering expertise for public policymaking: The challenges of contestability.” In: *Australian Journal of Public Administration* 83.2, pp. 156–172.

Heinkelmann-Wild, Tim et al. (Feb. 2023). “Blame shifting and blame obfuscation: The blame avoidance effects of delegation in the European Union.” In: *European Journal of Political Research* 62.1, pp. 221–238.

Heinkelmann-Wild, Tim and Bernhard Zangl (Oct. 2020). “Multilevel blame games: Blame-shifting in the European Union.” In: *Governance* 33.4, pp. 953–969.

Hwang, T. J. et al. (2020). “Association between FDA and EMA expedited approval programs and therapeutic value of new medicines: retrospective cohort study.” In: *BMJ* 371, p. m3434.

Jansen, M. P. et al. (2017). “Fair processes for priority setting: Putting theory into practice: comment on "expanded HTA: Enhancing fairness and legitimacy".” In: *International journal of health policy and management* 6.1, p. 43.

Jensen, C. and M. B. Petersen (2017). “The deservingness heuristic and the politics of health care.” In: *American Journal of Political Science* 61.1, pp. 68–83.

Kahneman, Daniel (2003). “A Perspective on Judgment and Choice: Mapping Bounded Rationality.” In: *American Psychologist* 58.9, pp. 697–720.

Kahneman, Daniel and Amos Tversky (Mar. 1979). “Prospect Theory: An Analysis of Decision under Risk.” In: *Econometrica* 47.2, p. 263.

Kahneman, Daniel and Amos Tversky (1984). “Choices, Values, and Frames.” In: *American Psychologist* 39.4, pp. 341–350.

Kapiriri, L. and D. Razavi (2017). “How have systematic priority setting approaches influenced policy making? A synthesis of the current literature.” In: *Health Policy* 121.9, pp. 937–946.

Koop, Christel and Chris Hanretty (2018). “Political Independence, Accountability, and the Quality of Regulatory Decision-Making.” In: *Comparative Political Studies* 51.1, pp. 38–75.

Koszegi, Botond and Matthew Rabin (Nov. 2006). “A Model of Reference-Dependent Preferences.” In: *The Quarterly Journal of Economics* 121.4, pp. 1133–1165.

Koszegi, Botond and Matthew Rabin (Aug. 2007). “Reference-Dependent Risk Attitudes.” In: *American Economic Review* 97.4, pp. 1047–1073.

Laenen, Tijs (Dec. 2025). “Framing social policy preferences: A scoping review and research agenda.” In: *Open Research Europe* 5, p. 371.

Langvatn, Silje A. and Cathrine Holst (2022). “Expert accountability: What does it mean, why is it challenging—and is it what we need?” In: *European Journal of Risk Regulation* 13.4, pp. 743–768.

Loupas, M. A., K. Athanasakis, and D. Zavras (2026). “Willingness to Pay per QALY: A Systematic Review of Demand-Side Valuations with a Focus on Age and Disease Severity.” In: *Appl Health Econ Health Policy* 24, pp. 47–63.

MacLeod, T. E., A. H. Harris, and A. Mahal (2016). “Stated and revealed preferences for funding new high-cost cancer drugs: a critical review of the evidence from patients, the public and payers.” In: *Patient-Patient-Centered Outcomes Res.* 9, pp. 201–222.

Maguire, Allegra, Emil Persson, and Gustav Tinghög (Dec. 2023). “Opportunity cost neglect: a meta-analysis.” In: *Journal of the Economic Science Association* 9.2, pp. 176–192.

Mahdiani, H., N. Münch, and N. W. Paul (2024). “A QALY is [still] a QALY is [still] a QALY?” In: *BMC Med Ethics* 25, p. 35.

McKie, J. and J. Richardson (2003). “The rule of rescue.” In: *Social science & medicine* 56.12, pp. 2407–2419.

McKie, John and Jeff Richardson (June 2003). “The Rule of Rescue.” In: *Social Science & Medicine* 56.12, pp. 2407–2419.

Mitton, C. R. et al. (2006). “Centralized drug review processes: Are they fair?” In: *Soc Sci Med.* 63.1, pp. 200–211.

Mor, J. et al. (2024). “Pharmaceutical industry promotional activities on social media: a scoping review.” In: *Journal of Pharmaceutical Health Services Research* 15.4, rmae022.

Morrell, L., S. Wordsworth, S. Rees, et al. (2017). “Does the Public Prefer Health Gain for Cancer Patients? A Systematic Review of Public Views on Cancer and its Characteristics.” In: *PharmacoEconomics* 35, pp. 793–804.

Mulinari, S. et al. (2020). “Five years of pharmaceutical industry funding of patient organisations in Sweden: Cross-sectional study of companies, patient organisations and drugs.” In: *PLOS ONE* 15.6, e0235021.

O’Donnell, Owen Andrew et al. (2008). *Analyzing health equity using household survey data: a guide to techniques and their implementation*. Washington, DC: World Bank Group.

Oliver, A. (2009). “A Fair Test of the Fair Innings?” In: *Medical Decision Making* 29.4, pp. 491–499.

Olofsson, S., U. G. Gerdtham, L. Hultkrantz, et al. (2018). “Measuring the end-of-life premium in cancer using individual ex ante willingness to pay.” In: *Eur J Health Econ* 19, pp. 807–820.

Parker, Lisa et al. (2019). “Asset exchange—interactions between patient groups and pharmaceutical industry: Australian qualitative study.” In: *BMJ* 367, p. l6694.

Pinho, M. and E. Dias Costa (2020). “Can mass media be an obstacle to rationing decisions? A case report from Portugal.” In: *Int J Health Gov.* 25.1, pp. 3–11.

Robb, K. A., A. E. Simon, A. Miles, et al. (2014). “Public perceptions of cancer: a qualitative study of the balance of positive and negative beliefs.” In: *BMJ Open* 4, e005434.

Rubagumya, F. et al. (2023). “Media coverage of cancer therapeutics: a review of literature.” In: *Journal of cancer policy* 36, p. 100418.

Schomerus, G., H. Matschinger, and M. Angermeyer (2006). “Preferences of the public regarding cutbacks in expenditure for patient care.” In: *Soc Psychiat Epidemiol* 41, pp. 369–377.

Shah, K. et al. (2015). *Do respondents completing abstract, hypothetical priority-setting exercises agree with the policy implications of their choices?* Office for Health Economics.

Sinclair, S. (2022). “Explaining rule of rescue obligations in healthcare allocation: allowing the patient to tell the right kind of story about their life.” In: *Medicine, Health Care and Philosophy* 25.1, pp. 31–46.

Sinclair, Sean (Mar. 2022). “Explaining rule of rescue obligations in healthcare allocation: allowing the patient to tell the right kind of story about their life.” In: *Medicine, Health Care and Philosophy* 25.1, pp. 31–46.

Singh, J. et al. (2012). “Does responsibility affect the public’s valuation of health care interventions? A relative valuation approach to health care safety.” In: *Value in Health* 15.5, pp. 690–698.

Steinbrook, Robert (2020). “Lobbying Expenditures, Campaign Contributions, and Health Care—Follow the Money.” In: *JAMA Internal Medicine* 180.3, pp. 467–469.

Thatcher, Mark and Alec Stone Sweet (Jan. 2002). “Theory and Practice of Delegation to Non-Majoritarian Institutions.” In: *West European Politics* 25.1, pp. 1–22.

Tversky, Amos and Daniel Kahneman (1973). “Availability: A heuristic for judging frequency and probability.” In: *Cognitive Psychology* 5.2, pp. 207–232.

Van Der Aa, M. et al. (2017). “Healthcare deservingness opinions of the general public and policymakers compared: A discrete choice experiment.” In: *The Social Legitimacy of Targeted Welfare*. In: Van Oorschot et al. (eds). Edward Elgar Publishing, pp. 241–260.

Vella Bonanno, P., M. Ermisch, B. Godman, et al. (2017). “Adaptive Pathways: Possible Next Steps for Payers in Preparation for Their Potential Implementation.” In: *Front Pharmacol.* 8. Accessed October 14, 2023.

Vis, Barbara (Sept. 2011). “Prospect Theory and Political Decision Making.” In: *Political Studies Review* 9.3, pp. 334–343.

Watters, S. M. (2016). “Fair Innings as a Basis for Prioritization: An Empirical Perspective.” In: *Prioritization in Medicine*. In: Nagel, E. and Lauerer, M. (eds). Springer, Cham.

Whitty, J. A., E. Lancsar, K. Rixon, et al. (2014). “A Systematic Review of Stated Preference Studies Reporting Public Preferences for Healthcare Priority Setting.” In: *Patient* 7, pp. 365–386.

Whitty, Jennifer A. et al. (2014). "Harnessing the Potential to Quantify Public Preferences for Healthcare Priorities Through Citizens' Juries." In: *International Journal of Health Policy and Management* 3.2, pp. 57–62.

.... **Word count**: 6,671 ....

6 Appendix

6.1 Regression Model

Table A1: Framing Effects on Public Support for Funding Expensive Cancer Medicines

	Ordinal logit
Loss (rescue) frame (vs. control)	1.478***
	[1.252, 1.746]
Gains (stewardship) frame (vs. control)	0.762**
	[0.644, 0.901]
Male (vs. female)	0.953
	[0.830, 1.095]
Age	1.016***
	[1.012, 1.021]
Income: middle (vs. low)	0.858
	[0.719, 1.023]
Income: high (vs. low)	0.944
	[0.778, 1.146]
Income: other/unknown	0.821
	[0.652, 1.035]
Eligible for Kela reimbursement	0.977
	[0.829, 1.151]
Num.Obs.	2458

Notes: Entries report odds ratios from an ordinal logistic regression. The dependent variable measures respondents' preferred funding decision for a novel, high-cost cancer medicine, ordered from unconditional public funding to outright rejection. The key independent variable captures experimental framing of the decision context. Control variables include gender, age, income group, education, eligibility for Kela medicine reimbursement, and self-reported medicine expenditure. Models are estimated using survey weights. 90% confidence intervals in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.